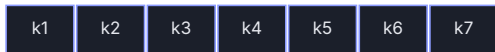


(a) no sync: launches are queued



CPU runs ahead, queue stays full

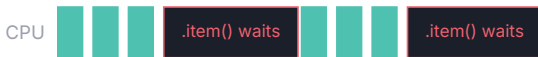
GPU stream 0



GPU stream 1

all-reduce (overlaps)

(b) `.item()` every step: the queue drains



GPU stream 0



GPU idle

time →